Name: Khattiya Wiriyakeattipong Student ID: 6730054721

HOMEWORK 5

Discrete-Time Fourier Transform

2110203 COMPUTER ENGINEERING MATHEMATICS II

Generative AI Usage Declaration

☐ **No AI Used:** I did not use any AI tools to complete this homework.

☑ **AI Assisted:** I used AI tools as detailed below:

Where	AI model	How
Question 2.2	Claude Opus 5	Verify the partial fraction calculation results.

Student Acknowledgement : I declare that I have reviewed, understood, and tested all AI-generated output. I take full responsibility for the accuracy, logic, and functionality of the final submitted work.

Signature: Khattiya. Date: 5 Sep 2026

2110203: COMP ENG MATH II (2026/1) Homework 5

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Write solutions and give explanations (1 point)

SEND ONLY PROBLEM 1. and 2.

1. (0.6 points, 0.2 points each) Use the properties of the discrete-time transform to determine $X(e^{j\omega})$ for the following sequences

1.1. $x[n] = (\frac{1}{3})^{|n|}$

$$x[n] = a^n u[n] + a^{-n} u[-n] - \delta[n] \quad ; \quad a = \frac{1}{3}$$

$$\left[a^n u[n] \leftrightarrow \frac{1}{1-ae^{-jw}} \right]$$

$$\left[a^{-n} u[-n] \leftrightarrow \frac{1}{1-ae^{jw}} \right]$$

$$\left[\delta[n] \leftrightarrow 1 \right]$$

$$; \; X(e^{jw}) = \frac{1}{1-ae^{-jw}} + \frac{1}{1-ae^{jw}} - 1$$

$$= \frac{(1-ae^{jw}) + (1-ae^{-jw})}{(1-ae^{-jw})(1-ae^{jw})} - 1$$

$$= \frac{2 - ae^{jw} - ae^{-jw}}{1 - \underbrace{\cancel{ae^{jw} - ae^{-jw}}}_{2acosw} + a^2} - 1$$

$$= \frac{2 - a(\underbrace{\cancel{e^{jw} + e^{-jw}}}_{2cosw})}{1 - 2acosw + a^2} - 1$$

$$= \frac{2 - 2acosw}{1 - 2acosw + a^2} - 1$$

$$= \frac{2 - 2acosw - (1 - 2acosw + a^2)}{1 - 2acosw + a^2}$$

$$= \frac{1 - a^2}{1 - 2acosw + a^2}$$

$$a = \frac{1}{3} \; ; \quad = \frac{1 - (1/3)^2}{1 - \frac{2}{3}cosw + (1/3)^2}$$

$$= \frac{4}{5 - 3cosw}$$

※.

1.2. $x[n] = a^n \cos(\Omega_0 n) \cdot u[n], |a| < 1$

$$= a^n \cdot \frac{1}{2}(e^{j\Omega_0 n} + e^{-j\Omega_0 n}) \cdot u[n]$$

$$= \frac{1}{2} e^{j\Omega_0 n} a^n u[n] + \frac{1}{2} e^{-j\Omega_0 n} a^n u[n]$$

let $G(e^{jw}) = \frac{1}{1-ae^{-jw}}$

$$\left[a^n u[n] \leftrightarrow \frac{1}{1-ae^{-jw}} \right]$$

$$\therefore X(e^{jw}) = \frac{1}{2} e^{j\Omega_0 n} G(e^{jw}) + \frac{1}{2} e^{-j\Omega_0 n} G(e^{jw})$$

$$= \frac{1}{2} G(e^{j(w-\Omega_0)}) + \frac{1}{2} G(e^{j(w+\Omega_0)})$$

$$= \frac{1/2}{1-ae^{-j(w-\Omega_0)}} + \frac{1/2}{1-ae^{-j(w+\Omega_0)}}$$

$$= \frac{1}{2}\left[\frac{1-ae^{-j(w+\Omega_0)} + 1-ae^{-j(w-\Omega_0)}}{(1-ae^{-j(w+\Omega_0)})(1-ae^{-j(w+\Omega_0)})} \right]$$

$$= \frac{1}{\cancel{2}}\left[\frac{\cancel{2} - a(\underbrace{\cancel{e^{-j(w-\Omega_0)} + e^{-j(w+\Omega_0)}}}_{2cos\Omega_0})}{1 - 2acos(\Omega_0)e^{-jw} + a^2 e^{-j2w}} \right]$$

$$= \frac{1 - acos(\Omega_0)e^{-jw}}{1 - 2acos(\Omega_0)e^{-jw} + a^2 e^{-j2w}}$$

※.

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1.3. $x[n] = (n+1)a^n \cdot u[n], |a| < 1$

$$= \underbrace{na^n \cdot u[n]}_{\text{I}} + \underbrace{a^n u[n]}_{\text{II}}$$

I ; $\left[a^n u[n] \leftrightarrow \frac{1}{1-ae^{-j\omega}} \right]$

let $G(e^{j\omega}) = \frac{1}{1-ae^{-j\omega}} = (1-ae^{-j\omega})^{-1}$

$$\therefore \frac{dG}{d\omega} = -(1-ae^{-j\omega})^{-2}(jae^{-j\omega})$$

$$= \frac{jae^{-j\omega}}{-(1-ae^{-j\omega})^2}$$

$\left[nx[n] \leftrightarrow j\frac{dX(e^{j\omega})}{d\omega} \right]$

$$\therefore nx[n] \leftrightarrow \frac{ae^{-j\omega}}{(1-ae^{-j\omega})^2}$$

II ; $\left[a^n u[n] \leftrightarrow \frac{1}{1-ae^{-j\omega}} \right]$

$$\therefore X(e^{j\omega}) = \frac{ae^{-j\omega}}{(1-ae^{-j\omega})^2} + \frac{1}{1-ae^{-j\omega}}$$

$$= \frac{ae^{-j\omega} + 1 - ae^{-j\omega}}{(1-ae^{-j\omega})^2}$$

$$= \frac{1}{(1-ae^{-j\omega})^2}$$

※

2. (0.4 points, 0.2 points each) Find the discrete-time sequence $x[n]$ with transforms in range $0 \le \omega \le 2\pi$ as follows

2.1.

$$X(e^{j\omega}) = -j\pi\delta\left(\omega - \frac{\pi}{3}\right) + \pi\delta\left(\omega - \frac{2\pi}{3}\right) + \pi\delta\left(\omega - \frac{4\pi}{3}\right) + j\pi\delta\left(\omega - \frac{5\pi}{3}\right)$$

$\left[e^{-j\omega_0 n} \leftrightarrow 2\pi\delta(\omega - \omega_0) \right]$

$$x[n] = \frac{1}{2\cancel{\pi}}\left[-j\cancel{\pi}e^{j\pi n/3} + \cancel{\pi}e^{j2\pi n/3} + \cancel{\pi}e^{j4\pi n/3} + j\cancel{\pi}e^{j5\pi n/3} \right]$$

$$= \frac{1}{2}\left[e^{j2\pi n/3} + e^{-j2\pi n/3} \right] + \frac{1}{2}\left[e^{-j\pi n/3} + e^{j\pi n/3} \right]$$

$$= \cos\left(\frac{2\pi n}{3}\right) + \sin\left(\frac{\pi n}{3}\right)$$

※.

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2.2.

$$X(e^{j\omega}) = \frac{1 - \frac{5}{6}e^{-j\omega}}{1 + \frac{1}{12}e^{-j\omega} - \frac{1}{12}e^{-j2\omega}} \times \frac{-12}{-12}$$

view e^{-jw} as z

$$X(z^{-1}) = \frac{-12 + 10z}{-12 - z + z^2} = \frac{2(-6+5z)}{(z-4)(z+3)} = \frac{\cancel{A}^{4}}{z-4} + \frac{\cancel{B}^{6}}{z+3}$$

When $z = 4$; $A = \frac{2(-6+5(4))}{4+3} = \frac{28}{7} = 4$

When $z = -3$; $B = \frac{2(-6+5(-3))}{-3-4} = \frac{-42}{-7} = 6$

$$\therefore X(z^{-1}) = \frac{4}{z-4} + \frac{6}{z+3}$$

$$= \frac{\cancel{4}^{1}}{-\cancel{4}\left(1 - \frac{z}{4}\right)} + \frac{\cancel{6}^{2}}{\cancel{3}\left(1 + \frac{z}{3}\right)}$$

$[z = e^{-jw}]$; $x[n] = -\left(\frac{1}{4}\right)^n u[n] + 2\left(-\frac{1}{3}\right)^n u[n]$ ※.

OPTIONAL, PRACTICE ONLY.

3. Using the DTFT, find the impulse response ($h[n]$) of the system governed by the difference equation

3.1. $y(n) = x[n] - 4x[n-1] + \frac{11}{12}y[n-1] + \frac{1}{12}y[n-2]$

3.2. $y(n) = x[n] - \frac{11}{15}y[n-1] + \frac{2}{15}y[n-2]$

and find output with $x[n] = (\frac{1}{2})^n u[n]$.